Classical information

Ex: $|a\rangle$ = "ket a" = column vector w/ 1 for "a" entry & 0 for all others.
AKA Standard basis vector

Ex: Bit vector $\{0, 1\}$

$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Ex: $\begin{pmatrix} 3/4 \\ 1/4 \end{pmatrix} = \frac{3}{4}|0\rangle + \frac{1}{4}|1\rangle$

Ex: Probabilistic state of heads & tails

$\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} = \frac{1}{2}|\text{heads}\rangle + \frac{1}{2}|\text{tails}\rangle$

$|\text{heads}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\quad |\text{tails}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Classical Operations

Matrix M $\quad M|a\rangle = |f(a)\rangle$

Ex:

a	$f_1(a)$
0	0
1	0

constant

a	$f_2(a)$
0	0
1	1

Identity

a	$f_3(a)$
0	1
1	0

NOT

a	$f_4(a)$
0	1
1	1

constant

$M|a\rangle = |f(a)\rangle$

$M_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ $\quad M_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\quad M_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\quad M_4 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$

$M_1|0\rangle = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle = |f_1(0)\rangle$

$M_1|1\rangle = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle = |f_1(1)\rangle$

$\langle a|$ = "bra a" = row vector

Ex: $\langle 0| = (1 \ \ 0)$ $\quad \langle 1| = (0 \ \ 1)$

$|b\rangle\langle a|$ = square matrix w/ 1 in entry of (b,a)

Ex: $|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix}(0 \ \ 1) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$2\times 1 \cdot 1\times 2$

$\begin{bmatrix} 1\cdot 0 & 1\cdot 1 \\ 0\cdot 0 & 0\cdot 1 \end{bmatrix}$

$M = \sum_{a \in \Sigma} |f(a)\rangle\langle a|$

Ex: f_4 $\quad \Sigma = \{0, 1\}$

$M_4 = |f_4(0)\rangle\langle 0| + |f_4(1)\rangle\langle 1| = |1\rangle\langle 0| + |1\rangle\langle 1|$

$$= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

$\langle a || b \rangle = \langle a | b \rangle \Rightarrow 1 \times 1$ matrix $\Rightarrow$ scaler/number

$$\langle a | b \rangle = \begin{cases} 1 & a = b \\ 0 & a \neq b \end{cases}$$

$$M|b\rangle = \left(\sum_{a \in \Sigma} |f(a)\rangle\langle a| \right) |b\rangle = \sum_{a \in \Sigma} |f(a)\rangle \langle a|b\rangle = |f(b)\rangle$$

Probabilistic operations & stochastic matrices

stochastic matrices:

- All entries are nonnegative real numbers
- The entries in every column sum to 1

columns all form probability vectors

Quantum Information

Quantum state vectors

Quantum state of a system represented by column vector.

- Entries are complex numbers.
- The sum of the absolute values squared of the entries is 1.
  $\Sigma |*|^2$

Euclidean norm

$$v = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \Rightarrow \|v\| = \sqrt{\sum_{k=1}^{n} |\alpha_k|^2}$$

Ex: $\begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$ $\quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$, $\quad \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$,

$$\begin{pmatrix} \frac{1+2i}{3} \\ -2/3 \end{pmatrix} = \frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle$$

$$\left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} + \frac{1}{2} = 1 \checkmark$$

$$\left|\frac{1+2i}{3}\right|^2 + \left|-\frac{2}{3}\right|^2 = \frac{1+4}{9} + \frac{4}{9} = \frac{5}{9} + \frac{4}{9} = \frac{9}{9} = 1 \checkmark$$

Plus state: $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

Minus state: $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$

Ex: $|\psi\rangle = \frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle = \begin{pmatrix} \frac{1+2i}{3} \\ -\frac{2}{3} \end{pmatrix}$,

$\langle 0|\psi\rangle = \frac{1+2i}{3}$ & $\langle 1|\psi\rangle = -\frac{2}{3}$

Ex: If $|\psi\rangle = \frac{1+2i}{3} - \frac{2}{3}|1\rangle$, then $\langle\psi| = \frac{1-2i}{3}\langle 0| - \frac{2}{3}\langle 1|$

$= \begin{pmatrix} \frac{1-2i}{3} & -\frac{2}{3} \end{pmatrix}$

Measuring quantum states

Standard Basis Measurement

Ex: measuring plus state

$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

$Pr(\text{outcome is } 0) = |\langle 0|+\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

$Pr(\text{outcome is } 1) = |\langle 1|+\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

Minus state: $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$

$Pr(\text{outcome is } 0) = |\langle 0|-\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

$Pr(\text{outcome is } 1) = |\langle 1|-\rangle|^2 = \left|\frac{-1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

Ex: $|\psi\rangle = \frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle$

$\Pr(\text{outcome is } 0) = |\langle 0|\psi\rangle|^2 = \left|\frac{1+2i}{3}\right|^2 = \frac{1+4}{9} = \frac{5}{9}$

$\Pr(\text{outcome is } 1) = |\langle 1|\psi\rangle|^2 = \left|-\frac{2}{3}\right|^2 = \frac{4}{9}$

Unitary Operations

Operations on quantum state vectors are represented by unitary matrices

Matrix U is unitary if: $UU^\dagger = I$, $U^\dagger U = I$

$U^{-1} = U^\dagger$ True for square matrices

Multiplying a unitary matrix to a quantum state vector results in another quantum state vector.

Unitary operation examples

1. Pauli operations:

$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

X — Bit flip/NOT

Y

Z — Phase flip

$X|0\rangle = |1\rangle$
$X|1\rangle = |0\rangle$

$Z|0\rangle = |0\rangle$
$Z|1\rangle = -|1\rangle$

2. Hadamard operation:

$$H = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

3. Phase operations:

$$P_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$ θ is a real number

$$S = P_{\pi/2} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \qquad T = P_{\pi/4} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$$

Action of Hadamard operation on some common qubit state vectors:

$$H|0\rangle = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2}+0 \\ 1/\sqrt{2}+0 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = |+\rangle$$

$$H|1\rangle = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 + 1/\sqrt{2} \\ 0 - 1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = |-\rangle$$

$$H|+\rangle = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}\begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/2 + 1/2 \\ 1/2 - 1/2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

$$H|-\rangle = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}\begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/2 - 1/2 \\ 1/2 + 1/2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

$$H|0\rangle = H|+\rangle \qquad H|+\rangle = |0\rangle$$

$$H|1\rangle = H|-\rangle \qquad H|-\rangle = H|1\rangle$$

Ex: $H\left(\frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle\right) = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}\begin{pmatrix} \frac{1+2i}{3} \\ -2/3 \end{pmatrix}$

$= \begin{pmatrix} \frac{-1+2i}{3\sqrt{2}} \\ \frac{3+2i}{3\sqrt{2}} \end{pmatrix} = \frac{-1+2i}{3\sqrt{2}}|0\rangle + \frac{3+2i}{3\sqrt{2}}|1\rangle$

Ex: $T|+\rangle = T\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) = \frac{1}{\sqrt{2}}T|0\rangle + \frac{1}{\sqrt{2}}T|1\rangle$

$= \frac{1}{\sqrt{2}}|0\rangle + \frac{1+i}{2}|1\rangle \qquad T|0\rangle = |0\rangle, \quad T|1\rangle = \frac{1+i}{\sqrt{2}}|1\rangle$

Ex: $H\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1+i}{2}|1\rangle\right) = \frac{1}{\sqrt{2}}H|0\rangle + \frac{1+i}{2}H|1\rangle$

$= \frac{1}{\sqrt{2}}|+\rangle + \frac{1+i}{2}|-\rangle = \left(\frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle\right) + \left(\frac{1+i}{2\sqrt{2}}|0\rangle - \frac{1+i}{2\sqrt{2}}|1\rangle\right)$

$= \left(\frac{1}{2} + \frac{1+i}{2\sqrt{2}}\right)|0\rangle + \left(\frac{1}{2} - \frac{1+i}{2\sqrt{2}}\right)|1\rangle$